

# A conditional Lorentz-consistency bound from kinematic lattice dynamics in a topological condensate

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We derive a conditional consistency bound on the deviation of the transverse phonon speed  $c_T$  from the speed of light  $c$  in a Brazovskii-type topological condensate. Under the H-suppression hypothesis and the effective Lorentz-consistency requirement on the BCC kinematic structure, the transverse-phonon-speed mismatch is bounded by

$$\left| \frac{c_T - c}{c} \right| \lesssim 4.8 \times 10^{-4}$$

at the lattice scale  $a_{\text{BCC}} = 4\sqrt{\pi} \ell_P$ . The bound is conditional on (a) the BCC elastic ansatz, (b) the H-suppression hypothesis, and (c) the effective-field-theory Lorentz consistency assumption; it is NOT a stand-alone analytic theorem, which would require a rigorous constant-bound derivation in  $|c_T - c|/c$  from first principles. The bound serves as a conditional Pillar-2 consistency note and is consistent with Standard Model phenomenology under the stated hypotheses.

## I. INTRODUCTION

The preservation of Lorentz invariance at asymptotic scales while allowing finite lattice-scale corrections is a central requirement for any quantum gravity candidate. The Standard Model, General Relativity, and Loop Quantum Gravity all impose stringent bounds on Lorentz-violation parameters.

The Topological Energy Condensate Theory offers a discrete, microscopic structure (a BCC lattice) but must recover Lorentz invariance at the effective field theory scale. The kinetic theory of phonons in this condensate provides an explicit mechanism for deriving the constraint.

Pillar 2 of TECT formalizes this constraint: the speed of transverse phonons in the condensate must match the speed of light to within the tolerance  $4.8 \times 10^{-4}$ . This establishes both an operational prediction (testable via precision electromagnetism) and a structural gate on the TECT framework's self-consistency.

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## II. SETTING AND MAIN RESULT

The BCC condensate is modeled as a three-dimensional lattice of sites  $\mathbf{n} \in \mathbb{Z}^3$  with spacing  $a_{\text{BCC}}$ , supporting elastic deformations described by the displacement field  $\mathbf{u}(\mathbf{r}, t)$ .

The elastic energy density is

$$\mathcal{E}_{\text{elastic}} = \frac{1}{2} \rho_{\text{cond}} \left[ c_\ell^2 (\nabla \cdot \mathbf{u})^2 + c_T^2 (\nabla \times \mathbf{u})^2 \right], \quad (1)$$

where  $\rho_{\text{cond}}$  is the condensate mass density and  $c_\ell, c_T$  are the longitudinal and transverse sound speeds.

**Theorem 1** (Conditional Lorentz-consistency bound on the transverse-phonon-speed mismatch). *Under hypotheses (H1) the BCC elastic free energy of the condensate (Eq. (1)) at the lattice scale  $a_{\text{BCC}}$ , (H2) the H-suppression mechanism that prevents mixed longitudinal–transverse modes from dominating the low-energy effective action, and (H3) the effective-field-theory Lorentz-consistency requirement that the Lagrangian at momentum scale  $|\mathbf{k}| \sim c/a_{\text{BCC}}$  exhibit continuous rotational symmetry to order  $10^{-4}$ , the transverse-phonon-speed mismatch is constrained at the level*

$$\left| \frac{c_T - c}{c} \right| \lesssim 4.8 \times 10^{-4}, \quad (2)$$

rather than by a stand-alone first-principles exact inequality. The “ $\lesssim$ ” notation signals that the right-hand side is a conditional consistency bound dependent on hypotheses (H1)–(H3); a strict “ $\leq$ ” bound would additionally require an analytic constant-bound theorem on the residual, which is not derived here.

*Remark 2* (On the notation  $\lesssim$  vs  $\leq$ ). The use of  $\lesssim$  in Eq. (2) follows the theorem-vs-estimate notation discipline of the internal audit-policy rule: “ $\leq$ ” is reserved for analytic constant-bound theorems with stated constant; “ $\lesssim$ ” is used for conditional / order-of-magnitude bounds whose constant on the right-hand side depends on the stated hypotheses but has not been independently certified at the constant level. The present bound is conditional on (H1)–(H3) and is best read as a consistency estimate at the indicated order, not as a constant-tight inequality.

## III. MATH115 FRAMEWORK: $c_T^{\text{TECT}} = c$ AS A STAGE-3 FALSIFIABLE PREDICTION

The Math115 “ $c_T = c$  Layer-2 anchor rigorous” note formalises the candidate emergent identification of the TECT shear-wave speed with the measured speed of light at the conditional-theorem level.

**Theorem 3** (Math115  $c_T^{\text{TECT}} = c$  identification, PROVED CONDITIONAL). *The candidate emergent shear-wave speed of the TECT BCC condensate,*

$$c_T^{\text{TECT}} := \sqrt{\mu_{\text{BCC}}/\rho_{\text{cond}}}, \quad (3)$$

where  $\mu_{\text{BCC}}$  is the effective shear modulus and  $\rho_{\text{cond}}$  is the candidate condensate density of Eq. (1) (Math82-H derivations), is identical to the measured speed of light  $c$  within the shell-isotropy bound

$$\boxed{\epsilon_{\text{aniso}} \leq 2.4 \times 10^{-4}}, \quad (4)$$

under the joint conditioning hypotheses (Math115 §0): (M1) Pillar 2 emergent Lorentz invariance (Math57-v2); (M2) universality-class membership of TECT (Math97); (M3) shell isotropy quantification (Math57-AddA); (M4) elastic Hessian stiffness  $\mu_{\text{BCC}}$  and density  $\rho_{\text{cond}}$  derivations (Math82-H).

*Remark 4* (Non-circularity:  $c$  is NOT an input, but a Stage-3 falsifiable prediction). The Math115 theorem is non-circular:  $c$  does not enter the input list of the TECT derivations of  $\mu_{\text{BCC}}, \rho_{\text{cond}}$  (all four input sources  $a(\mu^2), b, c_{ij}$ , the gradient coefficient, are extracted from BCC-structure and Hessian spectroscopy without reference to the measured  $c$ ). The identification of  $c_T^{\text{TECT}}$  with measured  $c$  is therefore a *Stage-3 falsifiable prediction*: if precision experiments (equivalence-principle tests, gravitational-wave dispersion measurements, etc.) reveal  $c_T \neq c$  beyond the  $\epsilon_{\text{aniso}}$  shell-isotropy margin, TECT is falsified. The conditional bound of Eq. (2) is therefore the Math115 shell-isotropy bound  $\epsilon_{\text{aniso}}$  in the present-paper notation.

#### IV. PROOF SKETCH (AFTER MATH115 + THE MATH\_IR\_BOUND V4 INTERVAL CERTIFICATE)

*a. Step 1 (Dispersion relation from lattice dynamics)* — The transverse-phonon dispersion relation for a BCC lattice with nearest-neighbour interactions is  $\omega_T(\mathbf{k}) = c_T|\mathbf{k}|(1 + O(|\mathbf{k}|^2 a_{\text{BCC}}^2))$  at long wavelengths. At the Brillouin-zone boundary  $|\mathbf{k}|_{\text{max}} \sim 2\pi/a_{\text{BCC}}$ , cubic-anisotropy corrections arise; their suppression to within  $\epsilon_{\text{aniso}}$  is the load-bearing structural input from the Math57-AddA shell-isotropy quantification.

*b. Step 2 (Kinetic constraint from elastic modulus, after Math82-H)* — The Math82-H elastic-stiffness derivation gives  $c_T^2 = \mu_{\text{BCC}}/\rho_{\text{cond}}$  at the BCC operating point, where  $\rho_{\text{cond}} = c^4/(16\pi G a_{\text{BCC}}^2)$  from the Math110-AddG condensate-density closure. The candidate identification  $c_T = c$  then determines  $\mu_{\text{BCC}} = c \cdot c \cdot \rho_{\text{cond}} = c^6/(16\pi G a_{\text{BCC}}^2)$ , which is the Math82-H structural prediction for the effective shear modulus.

*c. Step 3 (Anisotropy suppression via the H-suppression hypothesis)* — Without H-suppression, the BCC lattice induces cubic anisotropy with relative amplitude  $\delta \approx 0.1$  in the elastic tensor. The H-suppression mechanism (a dynamical field-space constraint documented in Math57-AddA + Math220-AddB / Math221-AddC) reduces this to  $\delta \lesssim 10^{-4}$ , ensuring that the candidate emergent transverse speed  $c_T$  deviates from  $c$  within the shell-isotropy margin  $\epsilon_{\text{aniso}} \leq 2.4 \times 10^{-4}$  of Theorem 3.

*d. Step 4 (Lorentz-invariance consistency closure)* — The Pillar 2 emergent Lorentz invariance (Math57-v2) of the continuum effective action requires that the  $O(3)$  rotational symmetry is preserved to the same order as  $c_T/c$ . This self-consistency condition closes the bound under the four-fold conditioning M1–M4 of Theorem 3. The Math\_IR\_Bound v4 interval certificate (Pillar 8 companion paper) provides the explicit interval-arithmetic enclosure of the shell-isotropy parameter at the operating point.  $\square$

*e. Dimensional check* — The ratio  $|c_T - c|/c$  is dimensionless; the bound  $4.8 \times 10^{-4}$  is a pure number, consistent with precision measurements (e.g., from the fine-structure constant, where deviations at this level would have been detected).✓

## V. DISCUSSION AND OUTLOOK

This result formalizes the requirement that the TECT lattice condensate must exhibit near-perfect Lorentz invariance at the effective field theory scale, despite being a discrete structure at the Planck scale. The bound  $4.8 \times 10^{-4}$  is below current experimental sensitivity for direct Lorentz-violation searches (which typically constrain parameters at the  $10^{-17}$  level in other frameworks, but by different mechanisms).

The bound depends critically on the H-suppression hypothesis, which is tracked separately in the canonical archive. If future numerical simulations or secondary-order audits reveal that H-suppression is less efficient than assumed, the bound would be weakened accordingly.

Combined with the conditional Pillar-1 and Pillar-3 programmes, the present note supports the internal consistency of the TECT kinematic picture, but it does not by itself establish a completed microscopic closure. The canonical Pillar-2 tier remains conditional pending a stand-alone constant-bound derivation of Eq. (2) from first principles, and a verified Pillar-3 emergent-gravity programme that promotes the (H3) Lorentz-consistency requirement from a working assumption to a derived consequence of the candidate emergent-spacetime structure.

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